

LEON XU

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PORTFOLIO



lxu062.github.io

EDUCATION

University of Michigan, College of Engineering

Ann Arbor, MI

B.S.E. in Mechanical Engineering | Expected May 2029 | Cumulative GPA: 3.78 / 4.00

Intended Minor in Business, Ross School of Business (applying March 2027)

Relevant Coursework: Design & Manufacturing I, Solid Mechanics, Strength of Materials, Engineering Materials, Dynamics & Vibrations, Thermodynamics, Programming & Data Structures (C++)

EXPERIENCE

PERA Global — CAD / CAE Software Development and Application Intern

Summer 2026

Leading Chinese engineering-simulation (CAE) software company, developer of the PERA SIM suite — Shanghai, China

- Developed and shipped a CAD defeaturing module to production for the company's CAE simulation software, automating a manual preparation step that took hours on large assemblies and now runs in seconds.
- Designed the user interface for a lightweight internal CAD application, designing for first-time users rather than trained CAD operators.
- Ran thermal and structural simulations in ANSYS Mechanical and PERA SIM — thermal analysis of an electronics enclosure, structural analysis of robotic-arm components — using self-designed hardware as test models for the software.

ENGINEERING PROJECTS

6-DOF Robotic Arm — Automated 3D-Printer Tending · Independent

2026 – Present

- Designing a 6-degree-of-freedom robotic arm (~2 kg payload, 600 mm reach) that uses visual servoing as its primary control loop to automate loading and unloading of a 3D printer.
- Owning the full engineering pipeline end-to-end — requirements definition, CAD design, simulation, fabrication, integration, and test; CAD assembly and structural analysis complete, full link set prototyped in ABS, fully assembled and free-standing base to end effector.

3DPrintSim — Anisotropic FEA for FDM-Printed Parts · Independent

2025 – Present

- Built a custom FEA tool that models the anisotropic mechanical behavior of FDM prints — directional strength (XY vs. Z) and shell-versus-infill structure — to predict how orientation and print settings affect part strength.
- Engineered the tool end-to-end with AI-assisted development (Claude Code, Codex, Cursor), combining structural simulation with hands-on print expertise to inform orientation and infill choices.

Motorized Vehicle — World-Record Speed Attempt · Lead Designer & Fabricator

2024 – 2026

- Led design and fabrication of a go-kart-engine-powered vehicle (running rolling chassis complete) targeting the Guinness World Record for fastest motorized wheelie bin — current mark 55 mph (88 km/h).
- Prototyped novel reverse-engineering methods to bring complex real-world geometry into CAD — iPhone LiDAR scanning, Gaussian-splat photogrammetry, and AI mesh generation.

LEADERSHIP & ACTIVITIES

FRC Robotics — Team 503 “Frog Force” · Controls Group Lead

2022 – 2025

- Led the merged ~20-person pneumatics + electrical group on a ~100-student team, and operated the robot on the competition drive team (2024–25).
- Member of **FIRST Hall of Fame** Team 503 (2021 Championship Chairman's Award) — qualified for the FIRST World Championship all four seasons; also a 2024 District Event win and consecutive Autonomous Awards.
- Founded and led the team's 3D-Printing Group (30–40 part orders/season for ~7 regional FTC teams) and built a 9-week controls training curriculum that onboarded 10 new members; 200+ hrs/season.

College Design Teams & Activities · University of Michigan

2025 – 2026

- MASA: reverse-engineered and CAD-modeled an LGSE purge/pre-press subsystem from schematics. MRacing (Formula SAE): built a 3D-printed scale MR25 model for wind-tunnel aero testing. Also: U-M Rifle Team.

TECHNICAL SKILLS

CAD: Autodesk Fusion 360, Autodesk Inventor, SolidWorks (pursuing CSWA), Siemens NX, FreeCAD, PTC Creo; engineering drawings & GD&T

Simulation: ANSYS Mechanical; Fusion 360 & Inventor (structural + thermal/electronics cooling)

Fabrication & Test: 3D printing — 7+ yrs FDM (Ender 3, Bambu X1C); MIG & stick welding; lathe & mill; drilling, cutting, grinding

Programming: Python, C++, MATLAB (working knowledge); AI-assisted development (Claude Code, Codex, Cursor)